



9582 - Brace for Impact! The Aftermath of Airbursts in Gas Giant Atmospheres

Cycle: 5, Proposal Category: GO

INVESTIGATORS

<i>Name</i>	<i>Institution</i>
Prof. Leigh Fletcher (PI) (ESA Member)	University of Leicester
Prof. Imke de Pater (CoI)	University of California - Berkeley
Michael H. Wong (CoI) (US Admin CoI)	University of California - Berkeley
Dr. Glenn S. Orton (CoI)	Jet Propulsion Laboratory
Dr. Arrate Antunano (CoI) (ESA Member)	Universidad del Pais Vasco
Dr. Ricardo Hueso (CoI) (ESA Member)	Universidad del Pais Vasco
Prof. Agustin Sanchez-Lavega (CoI) (ESA Member)	Universidad del Pais Vasco
Mr. Simon Toogood (CoI) (ESA Member)	University of Leicester
Dr. Oliver King (CoI) (ESA Member)	University of Leicester
Dr. Michael Roman (CoI) (ESA Member)	University of Leicester
Dr. Thibault Cavalie (CoI) (ESA Member)	Universite de Bordeaux
Dr. Amy Simon (CoI)	NASA Goddard Space Flight Center
Dr. Heidi B. Hammel (CoI)	Space Science Institute
Dr. Ramanakumar Sankar (CoI)	University of California - Berkeley
Prof. Csaba Palotai (CoI)	Florida Institute of Technology

OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Jupiter Trigger - Epoch One				
	1	JupiterMIRI-Trigger	MIRI Medium Resolution Spectroscopy	(1) JUPITER-TARGET
	2	MIRI Background	MIRI Medium Resolution Spectroscopy	(4) JUPITER-BACKGROUND
	3	JupiterNIRSPEC-Trigg er	NIRSpec IFU Spectroscopy	(2) JUPITER-TARGET-NIRSPEC

Folder	Observation	Label	Observing Template	Science Target
	4	JupiterNIRSPEC-Trigg er	NIRSpec IFU Spectroscopy	(3) JUPITER-TARGET-LOC2-NIRSPEC
Jupiter Trigger - Epoch Two				
	5	JupiterMIRI-Trigger	MIRI Medium Resolution Spectroscopy	(1) JUPITER-TARGET
	6	MIRI Background	MIRI Medium Resolution Spectroscopy	(4) JUPITER-BACKGROUND
	7	JupiterNIRSPEC-Trigg er	NIRSpec IFU Spectroscopy	(2) JUPITER-TARGET-NIRSPEC
	8	JupiterNIRSPEC-Trigg er	NIRSpec IFU Spectroscopy	(3) JUPITER-TARGET-LOC2-NIRSPEC
Saturn Trigger - Epoch One				
	9	SaturnMIRI-Trigger	MIRI Medium Resolution Spectroscopy	(5) SATURN-TARGET
	10	MIRI Background	MIRI Medium Resolution Spectroscopy	(7) SATURN-BACKGROUND
	11	SaturnNIRSPEC-Trigg er	NIRSpec IFU Spectroscopy	(6) SATURN-TARGET-NIRSPEC
Saturn Trigger - Epoch Two				
	12	SaturnMIRI-Trigger	MIRI Medium Resolution Spectroscopy	(5) SATURN-TARGET
	13	MIRI Background	MIRI Medium Resolution Spectroscopy	(7) SATURN-BACKGROUND
	14	SaturnNIRSPEC-Trigg er	NIRSpec IFU Spectroscopy	(6) SATURN-TARGET-NIRSPEC

ABSTRACT

We propose a non-disruptive target of opportunity proposal to observe the aftermath of a giant impact on either Jupiter or Saturn. These are rare opportunities to glimpse planetary accretion in action, and the mechanisms responsible for "atmospheric pollution" of giant planets in our Solar System and beyond. Although major impact events are likely to occur within the foreseen lifetime of JWST (ten smaller impacts have been detected in the past decade), they remain impossible to predict in advance. If a dark debris scar is discovered (e.g., for >200m impactors on Jupiter), then this JWST trigger will provide spectroscopic maps of reflected sunlight (NIRSpec/IFU) and thermal emission (MIRI/MRS), allowing us to derive the 3D structure of temperatures, winds, chemical composition and aerosols associated with these spectacular events. The distribution of energy and chemicals (like ammonia) from the impact can reveal the impactor direction, penetration, and tensile strength; the reducing or oxidising chemistry (hydrocarbons, nitriles, and CO) can reveal the water content of the impactor; and the glowing silicate debris can reveal its origins and chemical make-up. This multi-cycle ToO will specifically target spectroscopy that cannot be obtained from Earth, and will be supported by ground-based observations tracking the evolving impact scar morphology. Only large impacts with detected debris fields will be considered for this JWST trigger. Long-term multi-cycle status is requested due to the rarity of these extreme events, and the challenge of capturing them when the planet is in the

JWST field of regard.

OBSERVING DESCRIPTION

Target of Opportunity Proposal to observe impact debris on Jupiter and Saturn. This is a non-disruptive proposal to capture the changes in temperature and chemistry of the target of interest. We request long-term multi-cycle status given the rarity of these events and the challenges of triggering within FoR windows.

* Observations have been designed for both planets, although the time request is based on the larger of the two requests. Note that this request is for a SINGLE trigger of EITHER Jupiter OR Saturn, so the total time request is smaller than that given in APT. Specifically:

> Jupiter Trigger One: 5.5 hours (with smart accounting applied).

> Jupiter Follow up: 5.5 hours.

> Saturn Trigger One: 4.3 hours (with smart accounting applied).

> Saturn Follow up: 4.3 hours.

Thus the total time requested is 11.1 hours (to cover the Jupiter trigger). Both planets have two epochs defined, in the hope that the impact can be acquired twice within 7-14 days, subject to the target remaining within the FoR.

NOTES:

1. NIRSPEC POINTINGS: As the NIRSPEC FOV is the smallest of the modes proposed here, the Jupiter trigger features two NIRSPEC observations ~10 degrees longitude apart, to ensure that we capture the full extent of any impact scar. This is not necessary for the smaller angular size of Saturn.

2. DITHERING: All observations use a 4-point dither for an extended source, but given that Jupiter and Saturn are rotating, software has been developed to allow us to properly coadd the sequence of dithers and integrations.

3. LOCATION: A dummy location has been specified for the purposes of submission, precise targetting will be provided at the time of the trigger.

Note that we request the impact/storm location to be observed close to the central meridian, so a negative source-target separation constraint has been used to ensure that the target is nowhere near the planetary limb, while ensuring that we avoid direct-scheduling overheads.

4. BACKGROUND: A single MIRI background frame (an offset to sky, 180" away from Jupiter or Saturn) should be acquired to characterise any anomalous slopes or contributions. This should be executed as near as possible to the science frames, and therefore has been grouped as non-interruptable. Given the brightness of Jupiter and Saturn, we do not foresee the need for background observations with NIRSPEC.

5. GROUPS: As some regions of Jupiter's spectrum (particularly those beyond 11 μm) are likely to saturate in the second group, we have set `ngroups=5` (the minimum possible, with the expectation that the final group will be saturated and discarded) and used sufficient iterations to build up our signal-to-noise. For the brightest regions, this will allow us to subsample the groups to "desaturate" the Channel 3 data, as shown in the science justification.

Proposal 9582 - Targets - Brace for Impact! The Aftermath of Airbursts in Gas Giant Atmospheres

#	Name	Level 1	Level 2	Level 3
(1)	JUPITER-TARGET	STD=JUPITER	TYPE=PGRAPHIC, LONG=0, LAT=0	
Comments: Latitude and longitude will be specified at the time of the ToO trigger, and must be observed close to the central meridian. Extended=YES				
(2)	JUPITER-TARGET-NIRSPEC	STD=JUPITER	TYPE=PGRAPHIC, LONG=0, LAT=0	
Comments: Latitude and longitude will be specified at the time of the ToO trigger, and must be observed close to the central meridian. Extended=YES				
(3)	JUPITER-TARGET-LOC2-NIRSPEC	STD=JUPITER	TYPE=PGRAPHIC, LONG=10, LAT=0	
Comments: Latitude and longitude will be specified at the time of the ToO trigger, and must be observed close to the central meridian. Extended=YES				
(4)	JUPITER-BACKGROUND	STD=JUPITER	TYPE=POS_ANGLE, RAD=180, ANG=0, REF=NORTH	
Comments: Sky background for MIRI observations of Jupiter. Extended=YES				
(5)	SATURN-TARGET	STD=SATURN	TYPE=PGRAPHIC, LONG=0, LAT=0	
Comments: Latitude and longitude will be specified at the time of the ToO trigger, and must be observed close to the central meridian. Extended=YES				
(6)	SATURN-TARGET-NIRSPEC	STD=SATURN	TYPE=PGRAPHIC, LONG=0, LAT=0	
Comments: Latitude and longitude will be specified at the time of the ToO trigger, and must be observed close to the central meridian. Extended=YES				
(7)	SATURN-BACKGROUND	STD=SATURN	TYPE=POS_ANGLE, RAD=180, ANG=0, REF=NORTH	
Comments: Sky background for MIRI observations of Saturn. Extended=YES				

Proposal 9582 - Observation 1 - Brace for Impact! The Aftermath of Airbursts in Gas Giant Atmospheres

Observation	Proposal 9582, Observation 1: JupiterMIRI-Trigger												Mon Jun 08 20:01:04 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
	Background Observations:[MIRI Background (Obs 2)]												
Diagnos	Comments: This is a ToO trigger, but will trigger both epoch 1 and epoch 2 of the Jupiter observations to study the evolution of the impact debris. Carry-over status is requested, as impacts have a low probability of occurrence within a single cycle.												
	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
	(JupiterMIRI-Trigger (Obs 1)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.												
Solar System Targets	#	Name	Level 1				Level 2				Level 3		
	(1)	JUPITER-TARGET	STD=JUPITER				TYPE=PGRAPHIC, LONG=0, LAT=0						
	Comments: Latitude and longitude will be specified at the time of the ToO trigger, and must be observed close to the central meridian.												
	Extended=YES												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel				Simultaneous Imaging				Imager Subarray		Grating Wheel Direction	
	F1000W	All MRS				NO				FULL		Allow Auto Reorder	
Dithers	#	Dither Type					Optimized For				Direction		
	1	4-Point					EXTENDED SOURCE				NEGATIVE		
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	SHORT(A)	MRSLONG		FASTR1	5	8	1	Dither 1	4	32	521.708	
	1	SHORT(A)	MRSSHORT		FASTR1	5	8	1	Dither 1	4	32	521.708	
	2	MEDIUM(B)	MRSLONG		FASTR1	5	8	1	Dither 1	4	32	521.708	
	2	MEDIUM(B)	MRSSHORT		FASTR1	5	8	1	Dither 1	4	32	521.708	
	3	LONG(C)	MRSLONG		FASTR1	5	8	1	Dither 1	4	32	521.708	
	3	LONG(C)	MRSSHORT		FASTR1	5	8	1	Dither 1	4	32	521.708	

Proposal 9582 - Observation 1 - Brace for Impact! The Aftermath of Airbursts in Gas Giant Atmospheres

Special Requirements	Target Of Opportunity Response Time 14 Days to 28 Days, Carry-Over
	5 After 1 by 7 Days to 14 Days
	Group Observations 1, 2, 3, 4, Non-interruptible
	DEFAULT WINDOW: ANGULAR RATE JUPITER-TARGET FROM JWST LESS THAN 0.03
	DEFAULT WINDOW: NOT OCCULTATION OF JUPITER-TARGET BY JUPITER FROM JWST
	DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF JUPITER-TARGET BY IO FROM JWST
	DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF JUPITER-TARGET BY EUROPA FROM JWST
	DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF JUPITER-TARGET BY GANYMEDE FROM JWST
	DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF JUPITER-TARGET BY CALLISTO FROM JWST
	DEFAULT WINDOW: SEPARATION OF JUPITER-TARGET IO FROM JWST GREATER THAN 10"
	DEFAULT WINDOW: SEPARATION OF JUPITER-TARGET EUROPA FROM JWST GREATER THAN 10"
	DEFAULT WINDOW: SEPARATION OF JUPITER-TARGET GANYMEDE FROM JWST GREATER THAN 10"
	DEFAULT WINDOW: SEPARATION OF JUPITER-TARGET CALLISTO FROM JWST GREATER THAN 10"
	SEPARATION OF JUPITER-TARGET JUPITER FROM JWST LESS THAN -6"

Proposal 9582 - Observation 2 - Brace for Impact! The Aftermath of Airbursts in Gas Giant Atmospheres

Observation	Proposal 9582, Observation 2: MIRI Background												Mon Jun 08 20:01:04 GMT 2026	
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
	Background Observation For: [JupiterMIRI-Trigger (Obs 1)]													
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
	(MIRI Background (Obs 2)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.													
Solar System Targets	#	Name	Level 1					Level 2					Level 3	
	(4)	JUPITER-BACKGROUND	STD=JUPITER					TYPE=POS_ANGLE,RAD=180,ANG=0,REF=NORTH						
	Comments: Sky background for MIRI observations of Jupiter.													
	Extended=YES													
Acquisition	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray			Grating Wheel Direction		
		All MRS				NO			FULL			Allow Auto Reorder		
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	SHORT(A)	MRSLONG		FASTR1	5	8	1	None	1	8	130.427		
	1	SHORT(A)	MRSSHORT		FASTR1	5	8	1	None	1	8	130.427		
	2	MEDIUM(B)	MRSLONG		FASTR1	5	8	1	None	1	8	130.427		
	2	MEDIUM(B)	MRSSHORT		FASTR1	5	8	1	None	1	8	130.427		
	3	LONG(C)	MRSLONG		FASTR1	5	8	1	None	1	8	130.427		
	3	LONG(C)	MRSSHORT		FASTR1	5	8	1	None	1	8	130.427		

Proposal 9582 - Observation 2 - Brace for Impact! The Aftermath of Airbursts in Gas Giant Atmospheres

Special Requirements

Group Observations 1, 2, 3, 4, Non-interruptible

DEFAULT WINDOW: NOT OCCULTATION OF JUPITER-BACKGROUND BY JUPITER FROM JWST
DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF JUPITER-BACKGROUND BY IO FROM JWST
DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF JUPITER-BACKGROUND BY EUROPA FROM JWST
DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF JUPITER-BACKGROUND BY GANYMEDE FROM JWST
DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF JUPITER-BACKGROUND BY CALLISTO FROM JWST
DEFAULT WINDOW: SEPARATION OF JUPITER-BACKGROUND IO FROM JWST GREATER THAN 10"
DEFAULT WINDOW: SEPARATION OF JUPITER-BACKGROUND EUROPA FROM JWST GREATER THAN 10"
DEFAULT WINDOW: SEPARATION OF JUPITER-BACKGROUND GANYMEDE FROM JWST GREATER THAN 10"
DEFAULT WINDOW: SEPARATION OF JUPITER-BACKGROUND CALLISTO FROM JWST GREATER THAN 10"
DEFAULT WINDOW: ANGULAR RATE JUPITER-BACKGROUND FROM JWST LESS THAN 0.075

Proposal 9582 - Observation 3 - Brace for Impact! The Aftermath of Airbursts in Gas Giant Atmospheres

Observation	Proposal 9582, Observation 3: JupiterNIRSPEC-Trigger												Mon Jun 08 20:01:04 GMT 2026
	Diagnostic Status: Warning Observing Template: NIRSpec IFU Spectroscopy												
Diagnostics	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run. (JupiterNIRSPEC-Trigger (Obs 3)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.												
Solar System Targets	#	Name	Level 1				Level 2				Level 3		
	(2)	JUPITER-TARGET-NIRSPEC	STD=JUPITER				TYPE=PGRAPHIC, LONG=0, LAT=0						
	Comments: Latitude and longitude will be specified at the time of the ToO trigger, and must be observed close to the central meridian. Extended=YES												
Template	TA Method						HFF Readout Mode						
	NONE						false						
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points			
	1	4-POINT-DITHER											
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	G235H/F170LP	NRSRAPID	2	2	false	true	NONE	4	8	257.682	89558	
	2	G395H/F290LP	NRSRAPID	2	10	false	true	NONE	4	40	1288.412	89558	
Special Requirements	Group Observations 1, 2, 3, 4, Non-interruptible SEPARATION OF JUPITER-TARGET JUPITER FROM JWST LESS THAN -6" DEFAULT WINDOW: NOT OCCULTATION OF JUPITER-TARGET-NIRSPEC BY JUPITER FROM JWST DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF JUPITER-TARGET-NIRSPEC BY IO FROM JWST DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF JUPITER-TARGET-NIRSPEC BY EUROPA FROM JWST DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF JUPITER-TARGET-NIRSPEC BY GANYMEDE FROM JWST DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF JUPITER-TARGET-NIRSPEC BY CALLISTO FROM JWST DEFAULT WINDOW: SEPARATION OF JUPITER-TARGET-NIRSPEC IO FROM JWST GREATER THAN 10" DEFAULT WINDOW: SEPARATION OF JUPITER-TARGET-NIRSPEC EUROPA FROM JWST GREATER THAN 10" DEFAULT WINDOW: SEPARATION OF JUPITER-TARGET-NIRSPEC GANYMEDE FROM JWST GREATER THAN 10" DEFAULT WINDOW: SEPARATION OF JUPITER-TARGET-NIRSPEC CALLISTO FROM JWST GREATER THAN 10" DEFAULT WINDOW: ANGULAR RATE JUPITER-TARGET-NIRSPEC FROM JWST LESS THAN 0.075												

Proposal 9582 - Observation 4 - Brace for Impact! The Aftermath of Airbursts in Gas Giant Atmospheres

Observation	Proposal 9582, Observation 4: JupiterNIRSPEC-Trigger												Mon Jun 08 20:01:04 GMT 2026
	Diagnostic Status: Warning Observing Template: NIRSpec IFU Spectroscopy												
Diagnostics	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run. (JupiterNIRSPEC-Trigger (Obs 4)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.												
Solar System Targets	#	Name	Level 1				Level 2				Level 3		
	(3)	JUPITER-TARGET-LOC2-NIRSPEC	STD=JUPITER				TYPE=PGRAPHIC, LONG=10, LAT=0						
	Comments: Latitude and longitude will be specified at the time of the ToO trigger, and must be observed close to the central meridian. Extended=YES												
Template	TA Method						HFF Readout Mode						
	NONE						false						
Dithers	#	Dither Type			Size		Starting Point		Number of Points		Points		
	1	4-POINT-DITHER											
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	G235H/F170LP	NRSRAPID	2	2	false	true	NONE	4	8	257.682	89558	
	2	G395H/F290LP	NRSRAPID	2	10	false	true	NONE	4	40	1288.412	89558	
Special Requirements	Group Observations 1, 2, 3, 4, Non-interruptible SEPARATION OF JUPITER-TARGET-LOC2-NIRSPEC JUPITER FROM JWST LESS THAN -6" DEFAULT WINDOW: NOT OCCULTATION OF JUPITER-TARGET-LOC2-NIRSPEC BY JUPITER FROM JWST DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF JUPITER-TARGET-LOC2-NIRSPEC BY IO FROM JWST DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF JUPITER-TARGET-LOC2-NIRSPEC BY EUROPA FROM JWST DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF JUPITER-TARGET-LOC2-NIRSPEC BY GANYMEDE FROM JWST DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF JUPITER-TARGET-LOC2-NIRSPEC BY CALLISTO FROM JWST DEFAULT WINDOW: SEPARATION OF JUPITER-TARGET-LOC2-NIRSPEC IO FROM JWST GREATER THAN 10" DEFAULT WINDOW: SEPARATION OF JUPITER-TARGET-LOC2-NIRSPEC EUROPA FROM JWST GREATER THAN 10" DEFAULT WINDOW: SEPARATION OF JUPITER-TARGET-LOC2-NIRSPEC GANYMEDE FROM JWST GREATER THAN 10" DEFAULT WINDOW: SEPARATION OF JUPITER-TARGET-LOC2-NIRSPEC CALLISTO FROM JWST GREATER THAN 10" DEFAULT WINDOW: ANGULAR RATE JUPITER-TARGET-LOC2-NIRSPEC FROM JWST LESS THAN 0.075												

Proposal 9582 - Observation 5 - Brace for Impact! The Aftermath of Airbursts in Gas Giant Atmospheres

Observation	Proposal 9582, Observation 5: JupiterMIRI-Trigger												Mon Jun 08 20:01:04 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
	Background Observations:[MIRI Background (Obs 6)]												
Diagnostics	(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
	(JupiterMIRI-Trigger (Obs 5)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.												
Solar System Targets	#	Name	Level 1				Level 2				Level 3		
	(1)	JUPITER-TARGET	STD=JUPITER				TYPE=PGRAPHIC, LONG=0, LAT=0						
	Comments: Latitude and longitude will be specified at the time of the ToO trigger, and must be observed close to the central meridian.												
	Extended=YES												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel				Simultaneous Imaging				Imager Subarray		Grating Wheel Direction	
	F1000W	All MRS				NO				FULL		Allow Auto Reorder	
Dithers	#	Dither Type				Optimized For				Direction			
	1	4-Point				EXTENDED SOURCE				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	SHORT(A)	MRSLONG		FASTR1	5	8	1	Dither 1	4	32	521.708	
	1	SHORT(A)	MRSSHORT		FASTR1	5	8	1	Dither 1	4	32	521.708	
	2	MEDIUM(B)	MRSLONG		FASTR1	5	8	1	Dither 1	4	32	521.708	
	2	MEDIUM(B)	MRSSHORT		FASTR1	5	8	1	Dither 1	4	32	521.708	
	3	LONG(C)	MRSLONG		FASTR1	5	8	1	Dither 1	4	32	521.708	
	3	LONG(C)	MRSSHORT		FASTR1	5	8	1	Dither 1	4	32	521.708	

Proposal 9582 - Observation 5 - Brace for Impact! The Aftermath of Airbursts in Gas Giant Atmospheres

Special Requirements	5 After 1 by 7 Days to 14 Days
	Group Observations 5, 6, 7, 8, Non-interruptible
	DEFAULT WINDOW: ANGULAR RATE JUPITER-TARGET FROM JWST LESS THAN 0.03
	DEFAULT WINDOW: NOT OCCULTATION OF JUPITER-TARGET BY JUPITER FROM JWST
	DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF JUPITER-TARGET BY IO FROM JWST
	DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF JUPITER-TARGET BY EUROPA FROM JWST
	DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF JUPITER-TARGET BY GANYMEDE FROM JWST
	DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF JUPITER-TARGET BY CALLISTO FROM JWST
	DEFAULT WINDOW: SEPARATION OF JUPITER-TARGET IO FROM JWST GREATER THAN 10"
	DEFAULT WINDOW: SEPARATION OF JUPITER-TARGET EUROPA FROM JWST GREATER THAN 10"
	DEFAULT WINDOW: SEPARATION OF JUPITER-TARGET GANYMEDE FROM JWST GREATER THAN 10"
	DEFAULT WINDOW: SEPARATION OF JUPITER-TARGET CALLISTO FROM JWST GREATER THAN 10"
	SEPARATION OF JUPITER-TARGET JUPITER FROM JWST LESS THAN -6"

Proposal 9582 - Observation 6 - Brace for Impact! The Aftermath of Airbursts in Gas Giant Atmospheres

Observation	Proposal 9582, Observation 6: MIRI Background												Mon Jun 08 20:01:04 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
	Background Observation For: [JupiterMIRI-Trigger (Obs 5)]												
Diagnostics	(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
	(MIRI Background (Obs 6)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.												
Solar System Targets	#	Name	Level 1					Level 2				Level 3	
	(4)	JUPITER-BACKGROUND	STD=JUPITER					TYPE=POS_ANGLE,RAD=180,ANG=0,REF=NORTH					
	H												
	Comments: Sky background for MIRI observations of Jupiter. Extended=YES												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray			Grating Wheel Direction	
		All MRS				NO			FULL			Allow Auto Reorder	
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	SHORT(A)	MRSLONG		FASTR1	5	8	1	None	1	8	130.427	
	1	SHORT(A)	MRSSHORT		FASTR1	5	8	1	None	1	8	130.427	
	2	MEDIUM(B)	MRSLONG		FASTR1	5	8	1	None	1	8	130.427	
	2	MEDIUM(B)	MRSSHORT		FASTR1	5	8	1	None	1	8	130.427	
	3	LONG(C)	MRSLONG		FASTR1	5	8	1	None	1	8	130.427	
	3	LONG(C)	MRSSHORT		FASTR1	5	8	1	None	1	8	130.427	

Proposal 9582 - Observation 6 - Brace for Impact! The Aftermath of Airbursts in Gas Giant Atmospheres

Special Requirements	Group Observations 5, 6, 7, 8, Non-interruptible
	DEFAULT WINDOW: NOT OCCULTATION OF JUPITER-BACKGROUND BY JUPITER FROM JWST
	DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF JUPITER-BACKGROUND BY IO FROM JWST
	DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF JUPITER-BACKGROUND BY EUROPA FROM JWST
	DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF JUPITER-BACKGROUND BY GANYMEDE FROM JWST
	DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF JUPITER-BACKGROUND BY CALLISTO FROM JWST
	DEFAULT WINDOW: SEPARATION OF JUPITER-BACKGROUND IO FROM JWST GREATER THAN 10"
	DEFAULT WINDOW: SEPARATION OF JUPITER-BACKGROUND EUROPA FROM JWST GREATER THAN 10"
	DEFAULT WINDOW: SEPARATION OF JUPITER-BACKGROUND GANYMEDE FROM JWST GREATER THAN 10"
	DEFAULT WINDOW: SEPARATION OF JUPITER-BACKGROUND CALLISTO FROM JWST GREATER THAN 10"
	DEFAULT WINDOW: ANGULAR RATE JUPITER-BACKGROUND FROM JWST LESS THAN 0.075

Proposal 9582 - Observation 7 - Brace for Impact! The Aftermath of Airbursts in Gas Giant Atmospheres

Observation	Proposal 9582, Observation 7: JupiterNIRSPEC-Trigger												Mon Jun 08 20:01:04 GMT 2026
	Diagnostic Status: Warning Observing Template: NIRSpec IFU Spectroscopy												
Diagnostics	(Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run. (JupiterNIRSPEC-Trigger (Obs 7)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.												
Solar System Targets	#	Name	Level 1				Level 2				Level 3		
	(2)	JUPITER-TARGET-NIRSPEC	STD=JUPITER				TYPE=PGRAPHIC, LONG=0, LAT=0						
	Comments: Latitude and longitude will be specified at the time of the ToO trigger, and must be observed close to the central meridian. Extended=YES												
Template	TA Method						HFF Readout Mode						
	NONE						false						
Dithers	#	Dither Type			Size		Starting Point		Number of Points		Points		
	1	4-POINT-DITHER											
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	G235H/F170LP	NRSRAPID	2	2	false	true	NONE	4	8	257.682		
	2	G395H/F290LP	NRSRAPID	2	10	false	true	NONE	4	40	1288.412		
Special Requirements	Group Observations 5, 6, 7, 8, Non-interruptible SEPARATION OF JUPITER-TARGET JUPITER FROM JWST LESS THAN -6" DEFAULT WINDOW: NOT OCCULTATION OF JUPITER-TARGET-NIRSPEC BY JUPITER FROM JWST DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF JUPITER-TARGET-NIRSPEC BY IO FROM JWST DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF JUPITER-TARGET-NIRSPEC BY EUROPA FROM JWST DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF JUPITER-TARGET-NIRSPEC BY GANYMEDE FROM JWST DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF JUPITER-TARGET-NIRSPEC BY CALLISTO FROM JWST DEFAULT WINDOW: SEPARATION OF JUPITER-TARGET-NIRSPEC IO FROM JWST GREATER THAN 10" DEFAULT WINDOW: SEPARATION OF JUPITER-TARGET-NIRSPEC EUROPA FROM JWST GREATER THAN 10" DEFAULT WINDOW: SEPARATION OF JUPITER-TARGET-NIRSPEC GANYMEDE FROM JWST GREATER THAN 10" DEFAULT WINDOW: SEPARATION OF JUPITER-TARGET-NIRSPEC CALLISTO FROM JWST GREATER THAN 10" DEFAULT WINDOW: ANGULAR RATE JUPITER-TARGET-NIRSPEC FROM JWST LESS THAN 0.075												

Proposal 9582 - Observation 8 - Brace for Impact! The Aftermath of Airbursts in Gas Giant Atmospheres

Observation	Proposal 9582, Observation 8: JupiterNIRSPEC-Trigger												Mon Jun 08 20:01:04 GMT 2026
	Diagnostic Status: Warning Observing Template: NIRSpec IFU Spectroscopy												
Diagnostics	(Visit 8:1) Warning (Form): Overheads are provisional until the Visit Planner has been run. (JupiterNIRSPEC-Trigger (Obs 8)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.												
Solar System Targets	#	Name	Level 1				Level 2				Level 3		
	(3)	JUPITER-TARGET-LOC2-NIRSPEC	STD=JUPITER				TYPE=PGRAPHIC, LONG=10, LAT=0						
	Comments: Latitude and longitude will be specified at the time of the ToO trigger, and must be observed close to the central meridian. Extended=YES												
Template	TA Method						HFF Readout Mode						
	NONE						false						
Dithers	#	Dither Type			Size		Starting Point		Number of Points		Points		
	1	4-POINT-DITHER											
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	G235H/F170LP	NRSRAPID	2	2	false	true	NONE	4	8	257.682		
	2	G395H/F290LP	NRSRAPID	2	10	false	true	NONE	4	40	1288.412		
Special Requirements	Group Observations 5, 6, 7, 8, Non-interruptible												
	SEPARATION OF JUPITER-TARGET-LOC2-NIRSPEC JUPITER FROM JWST LESS THAN -6" DEFAULT WINDOW: NOT OCCULTATION OF JUPITER-TARGET-LOC2-NIRSPEC BY JUPITER FROM JWST DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF JUPITER-TARGET-LOC2-NIRSPEC BY IO FROM JWST DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF JUPITER-TARGET-LOC2-NIRSPEC BY EUROPA FROM JWST DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF JUPITER-TARGET-LOC2-NIRSPEC BY GANYMEDE FROM JWST DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF JUPITER-TARGET-LOC2-NIRSPEC BY CALLISTO FROM JWST DEFAULT WINDOW: SEPARATION OF JUPITER-TARGET-LOC2-NIRSPEC IO FROM JWST GREATER THAN 10" DEFAULT WINDOW: SEPARATION OF JUPITER-TARGET-LOC2-NIRSPEC EUROPA FROM JWST GREATER THAN 10" DEFAULT WINDOW: SEPARATION OF JUPITER-TARGET-LOC2-NIRSPEC GANYMEDE FROM JWST GREATER THAN 10" DEFAULT WINDOW: SEPARATION OF JUPITER-TARGET-LOC2-NIRSPEC CALLISTO FROM JWST GREATER THAN 10" DEFAULT WINDOW: ANGULAR RATE JUPITER-TARGET-LOC2-NIRSPEC FROM JWST LESS THAN 0.075												

Proposal 9582 - Observation 9 - Brace for Impact! The Aftermath of Airbursts in Gas Giant Atmospheres

Observation	Proposal 9582, Observation 9: SaturnMIRI-Trigger												Mon Jun 08 20:01:04 GMT 2026	
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
	Background Observations:[MIRI Background (Obs 10)]													
	Comments: This is a ToO trigger, but will trigger both epoch 1 and epoch 2 of the Saturn observations to study the evolution of the impact debris. Carry-over status is requested, as impacts have a low probability of occurrence within a single cycle.													
Diagnostics	(Visit 9:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
	(SaturnMIRI-Trigger (Obs 9)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.													
Solar System Targets	#	Name	Level 1				Level 2				Level 3			
	(5)	SATURN-TARGET	STD=SATURN				TYPE=PGRAPHIC, LONG=0, LAT=0							
	Comments: Latitude and longitude will be specified at the time of the ToO trigger, and must be observed close to the central meridian. Extended=YES													
Acquisition	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel				Simultaneous Imaging				Imager Subarray		Grating Wheel Direction		
	F1000W	All MRS				NO				FULL		Allow Auto Reorder		
Dithers	#	Dither Type					Optimized For					Direction		
	1	4-Point					EXTENDED SOURCE					NEGATIVE		
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	SHORT(A)	MRSLONG		FASTR1	5	8	1	Dither 1	4	32	521.708		
	1	SHORT(A)	MRSSHORT		FASTR1	5	8	1	Dither 1	4	32	521.708		
	2	MEDIUM(B)	MRSLONG		FASTR1	5	8	1	Dither 1	4	32	521.708		
	2	MEDIUM(B)	MRSSHORT		FASTR1	5	8	1	Dither 1	4	32	521.708		
	3	LONG(C)	MRSLONG		FASTR1	5	8	1	Dither 1	4	32	521.708		
	3	LONG(C)	MRSSHORT		FASTR1	5	8	1	Dither 1	4	32	521.708		

Proposal 9582 - Observation 9 - Brace for Impact! The Aftermath of Airbursts in Gas Giant Atmospheres

Special Requirements	Target Of Opportunity Response Time 14 Days to 28 Days, Carry-Over
	12 After 9 by 7 Days to 14 Days
	Group Observations 9, 10, 11, Non-interruptible
	SEPARATION OF SATURN-TARGET SATURN FROM JWST LESS THAN -3"
	DEFAULT WINDOW: NOT OCCULTATION OF SATURN-TARGET BY SATURN FROM JWST
	DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF SATURN-TARGET BY TITAN FROM JWST
	DEFAULT WINDOW: SEPARATION OF SATURN-TARGET RHEA FROM JWST GREATER THAN 10"
	DEFAULT WINDOW: SEPARATION OF SATURN-TARGET TITAN FROM JWST GREATER THAN 10"
	DEFAULT WINDOW: ANGULAR RATE SATURN-TARGET FROM JWST LESS THAN 0.075

Proposal 9582 - Observation 10 - Brace for Impact! The Aftermath of Airbursts in Gas Giant Atmospheres

Observation	Proposal 9582, Observation 10: MIRI Background												Mon Jun 08 20:01:04 GMT 2026	
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
	Background Observation For: [SaturnMIRI-Trigger (Obs 9)]													
Diagnostics	(Visit 10:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
	(MIRI Background (Obs 10)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.													
Solar System Targets	#	Name	Level 1					Level 2					Level 3	
	(7)	SATURN-BACKGROUND	STD=SATURN					TYPE=POS_ANGLE,RAD=180,ANG=0,REF=NORTH						
	Comments: Sky background for MIRI observations of Saturn.													
	Extended=YES													
Acquisition	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray			Grating Wheel Direction		
	F1000W	All MRS				NO			FULL			Allow Auto Reorder		
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	SHORT(A)	MRSLONG		FASTR1	5	8	1	None	1	8	130.427		
	1	SHORT(A)	MRSSHORT		FASTR1	5	8	1	None	1	8	130.427		
	2	MEDIUM(B)	MRSLONG		FASTR1	5	8	1	None	1	8	130.427		
	2	MEDIUM(B)	MRSSHORT		FASTR1	5	8	1	None	1	8	130.427		
	3	LONG(C)	MRSLONG		FASTR1	5	8	1	None	1	8	130.427		
	3	LONG(C)	MRSSHORT		FASTR1	5	8	1	None	1	8	130.427		

Proposal 9582 - Observation 10 - Brace for Impact! The Aftermath of Airbursts in Gas Giant Atmospheres

Special Requirements	Group Observations 9, 10, 11, Non-interruptible DEFAULT WINDOW: NOT OCCULTATION OF SATURN-BACKGROUND BY SATURN FROM JWST DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF SATURN-BACKGROUND BY TITAN FROM JWST DEFAULT WINDOW: SEPARATION OF SATURN-BACKGROUND RHEA FROM JWST GREATER THAN 10" DEFAULT WINDOW: SEPARATION OF SATURN-BACKGROUND TITAN FROM JWST GREATER THAN 10" DEFAULT WINDOW: ANGULAR RATE SATURN-BACKGROUND FROM JWST LESS THAN 0.075
----------------------	--

Proposal 9582 - Observation 11 - Brace for Impact! The Aftermath of Airbursts in Gas Giant Atmospheres

Observation	Proposal 9582, Observation 11: SaturnNIRSPEC-Trigger											Mon Jun 08 20:01:04 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 11:1) Warning (Form): Overheads are provisional until the Visit Planner has been run. (SaturnNIRSPEC-Trigger (Obs 11)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.											
Solar System Targets	#	Name	Level 1				Level 2				Level 3	
	(6)	SATURN-TARGET-NIRSPEC	STD=SATURN				TYPE=PGRAPHIC, LONG=0, LAT=0					
	Comments: Latitude and longitude will be specified at the time of the ToO trigger, and must be observed close to the central meridian. Extended=YES											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type			Size		Starting Point			Number of Points		Points
	1	4-POINT-DITHER										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G235H/F170LP	NRSRAPID	2	5	false	true	NONE	4	20	644.206	89558
	2	G395H/F290LP	NRSRAPID	2	10	false	true	NONE	4	40	1288.412	89558
Special Requirements	Group Observations 9, 10, 11, Non-interruptible SEPARATION OF SATURN-TARGET SATURN FROM JWST LESS THAN -3" DEFAULT WINDOW: NOT OCCULTATION OF SATURN-TARGET-NIRSPEC BY SATURN FROM JWST DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF SATURN-TARGET-NIRSPEC BY TITAN FROM JWST DEFAULT WINDOW: SEPARATION OF SATURN-TARGET-NIRSPEC RHEA FROM JWST GREATER THAN 10" DEFAULT WINDOW: SEPARATION OF SATURN-TARGET-NIRSPEC TITAN FROM JWST GREATER THAN 10" DEFAULT WINDOW: ANGULAR RATE SATURN-TARGET-NIRSPEC FROM JWST LESS THAN 0.075											

Proposal 9582 - Observation 12 - Brace for Impact! The Aftermath of Airbursts in Gas Giant Atmospheres

Observation	Proposal 9582, Observation 12: SaturnMIRI-Trigger												Mon Jun 08 20:01:04 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
	Background Observations:[MIRI Background (Obs 13)]												
Diagnostics	(Visit 12:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
	(SaturnMIRI-Trigger (Obs 12)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.												
Solar System Targets	#	Name	Level 1					Level 2			Level 3		
	(5)	SATURN-TARGET	STD=SATURN					TYPE=PGRAPHIC,LONG=0,LAT=0					
	Comments: Latitude and longitude will be specified at the time of the ToO trigger, and must be observed close to the central meridian.												
	Extended=YES												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray		Grating Wheel Direction		
	F1000W	All MRS				NO			FULL		Allow Auto Reorder		
Dithers	#	Dither Type					Optimized For			Direction			
	1	4-Point					EXTENDED SOURCE			NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	SHORT(A)	MRSLONG		FASTR1	5	8	1	Dither 1	4	32	521.708	
	1	SHORT(A)	MRSSHORT		FASTR1	5	8	1	Dither 1	4	32	521.708	
	2	MEDIUM(B)	MRSLONG		FASTR1	5	8	1	Dither 1	4	32	521.708	
	2	MEDIUM(B)	MRSSHORT		FASTR1	5	8	1	Dither 1	4	32	521.708	
	3	LONG(C)	MRSLONG		FASTR1	5	8	1	Dither 1	4	32	521.708	
	3	LONG(C)	MRSSHORT		FASTR1	5	8	1	Dither 1	4	32	521.708	

Proposal 9582 - Observation 12 - Brace for Impact! The Aftermath of Airbursts in Gas Giant Atmospheres

Special Requirements

12 After 9 by 7 Days to 14 Days
Group Observations 12, 13, 14, Non-interruptible

SEPARATION OF SATURN-TARGET SATURN FROM JWST LESS THAN -3"
DEFAULT WINDOW: NOT OCCULTATION OF SATURN-TARGET BY SATURN FROM JWST
DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF SATURN-TARGET BY TITAN FROM JWST
DEFAULT WINDOW: SEPARATION OF SATURN-TARGET RHEA FROM JWST GREATER THAN 10"
DEFAULT WINDOW: SEPARATION OF SATURN-TARGET TITAN FROM JWST GREATER THAN 10"
DEFAULT WINDOW: ANGULAR RATE SATURN-TARGET FROM JWST LESS THAN 0.075

Proposal 9582 - Observation 13 - Brace for Impact! The Aftermath of Airbursts in Gas Giant Atmospheres

Observation	Proposal 9582, Observation 13: MIRI Background												Mon Jun 08 20:01:04 GMT 2026	
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
	Background Observation For: [SaturnMIRI-Trigger (Obs 12)]													
Diagnostics	(Visit 13:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
	(MIRI Background (Obs 13)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.													
Solar System Targets	#	Name	Level 1					Level 2					Level 3	
	(7)	SATURN-BACKGROUND	STD=SATURN					TYPE=POS_ANGLE,RAD=180,ANG=0,REF=NORTH						
	Comments: Sky background for MIRI observations of Saturn.													
	Extended=YES													
Acquisition	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray			Grating Wheel Direction		
	F1000W	All MRS				NO			FULL			Allow Auto Reorder		
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	SHORT(A)	MRSLONG		FASTR1	5	8	1	None	1	8	130.427		
	1	SHORT(A)	MRSSHORT		FASTR1	5	8	1	None	1	8	130.427		
	2	MEDIUM(B)	MRSLONG		FASTR1	5	8	1	None	1	8	130.427		
	2	MEDIUM(B)	MRSSHORT		FASTR1	5	8	1	None	1	8	130.427		
	3	LONG(C)	MRSLONG		FASTR1	5	8	1	None	1	8	130.427		
	3	LONG(C)	MRSSHORT		FASTR1	5	8	1	None	1	8	130.427		

Proposal 9582 - Observation 13 - Brace for Impact! The Aftermath of Airbursts in Gas Giant Atmospheres

Special Requirements

Group Observations 12, 13, 14, Non-interruptible

DEFAULT WINDOW: NOT OCCULTATION OF SATURN-BACKGROUND BY SATURN FROM JWST
DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF SATURN-BACKGROUND BY TITAN FROM JWST
DEFAULT WINDOW: SEPARATION OF SATURN-BACKGROUND RHEA FROM JWST GREATER THAN 10"
DEFAULT WINDOW: SEPARATION OF SATURN-BACKGROUND TITAN FROM JWST GREATER THAN 10"
DEFAULT WINDOW: ANGULAR RATE SATURN-BACKGROUND FROM JWST LESS THAN 0.075

Proposal 9582 - Observation 14 - Brace for Impact! The Aftermath of Airbursts in Gas Giant Atmospheres

Observation	Proposal 9582, Observation 14: SaturnNIRSPEC-Trigger											Mon Jun 08 20:01:04 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 14:1) Warning (Form): Overheads are provisional until the Visit Planner has been run. (SaturnNIRSPEC-Trigger (Obs 14)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.											
Solar System Targets	#	Name	Level 1				Level 2				Level 3	
	(6)	SATURN-TARGET-NIRSPEC	STD=SATURN				TYPE=PGRAPHIC, LONG=0, LAT=0					
	Comments: Latitude and longitude will be specified at the time of the ToO trigger, and must be observed close to the central meridian. Extended=YES											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type			Size		Starting Point		Number of Points		Points	
	1	4-POINT-DITHER										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G235H/F170LP	NRSRAPID	2	5	false	true	NONE	4	20	644.206	89558
	2	G395H/F290LP	NRSRAPID	2	10	false	true	NONE	4	40	1288.412	89558
Special Requirements	Group Observations 12, 13, 14, Non-interruptible SEPARATION OF SATURN-TARGET SATURN FROM JWST LESS THAN -3" DEFAULT WINDOW: NOT OCCULTATION OF SATURN-TARGET-NIRSPEC BY SATURN FROM JWST DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF SATURN-TARGET-NIRSPEC BY TITAN FROM JWST DEFAULT WINDOW: SEPARATION OF SATURN-TARGET-NIRSPEC RHEA FROM JWST GREATER THAN 10" DEFAULT WINDOW: SEPARATION OF SATURN-TARGET-NIRSPEC TITAN FROM JWST GREATER THAN 10" DEFAULT WINDOW: ANGULAR RATE SATURN-TARGET-NIRSPEC FROM JWST LESS THAN 0.075											